

HR EMPLOYEE ATTRITION ANALYSIS

Executive Insights & Recommendations | CRISP-DM

Key workforce findings, retention risks and recommended actions

Power BI HR Employee Attrition Analytics Portfolio Project

EXECUTIVE SUMMARY

Overall attrition is 23.8% (3,571 of 14,999 employees). The strongest patterns are lower satisfaction among leavers, substantially lower attrition among promoted employees, and higher attrition in lower salary bands. Workload also shows a non-linear relationship with attrition. Departmental performance is relatively consistent, suggesting performance alone does not explain turnover.

1. Business Understanding

Objective: Identify workforce patterns associated with employee attrition and provide practical retention recommendations.

- Measure overall attrition and identify departments with elevated turnover.
- Assess attrition in relation to satisfaction, workload, promotion and salary.
- Compare employee performance across departments.
- Translate findings into targeted actions for HR and management.

2. Data Understanding

- Source: Kaggle HR Employee dataset.
- 14,999 employee records and 11 workforce variables.
- 3,571 employees left and 11,428 stayed; overall attrition rate is 23.8%.
- Variables cover satisfaction, performance, projects, monthly hours, tenure, work accidents, promotion, department and salary.
- Data quality checks identified no missing values and no duplicate employee IDs.

3. Data Preparation

- Standardized field names and corrected data types.
- Cleaned department labels, including HR, R&D and Product Management.
- Created satisfaction and workload bands.
- Created readable attrition and promotion status categories.
- Created DAX measures for employee counts, attrition, satisfaction, performance, hours, projects and tenure.

4. Modeling / Analytical Approach

Method: Descriptive and diagnostic workforce analysis in Power BI using DAX measures and comparative visuals. No predictive model was developed; the objective was to identify decision-useful workforce patterns.

Analysis Area	Key Finding	Management Meaning
Satisfaction	Stayed: 66.7% average satisfaction; left: 44.0%. Low-satisfaction band has ~53.7% attrition.	Employee experience is a major retention risk.
Promotion	Attrition is ~24.2% for not promoted vs. ~5.9% for promoted employees.	Career progression is strongly associated with retention.
Salary	Attrition is ~29.7% low salary, ~20.4% medium and ~6.6% high.	Lower salary levels show substantially higher turnover.
Workload	Leavers average ~207 monthly hours vs. ~199 among those who stayed; very high hours show elevated attrition.	Monitor workload with context; the relationship is non-linear.
Departments	HR has the highest attrition rate (~29.1%); Sales has the largest departure volume. R&D and Management have lower attrition.	Retention actions should be department-specific.
Performance	Departmental average performance is relatively consistent (~0.70–0.72). Management combines high performance with low attrition.	Performance alone does not explain turnover.

5. Evaluation

Overall finding: Attrition is associated with multiple workforce factors rather than one isolated variable.

- Low satisfaction is strongly associated with higher attrition.
- Promotion status shows one of the clearest retention differences.
- Lower salary levels show substantially higher attrition.
- Workload is associated with attrition, but the relationship is non-linear.
- Departmental performance differences are relatively small compared with attrition differences.

Analytical caution: These findings show associations within the dataset and do not prove that any single factor causes employees to leave.

6. Deployment / Recommended Actions

1. Prioritize satisfaction: Run targeted engagement analysis in high-attrition departments and identify the specific drivers of dissatisfaction.

2. Strengthen career progression: Improve promotion transparency, internal mobility, mentoring and skills-development pathways.

3. Review lower salary bands: Assess compensation competitiveness and combine pay reviews with recognition and career opportunities.

4. Monitor workload: Track monthly hours, project allocation and workload concentration, especially in high-attrition teams.

5. Focus department interventions: Prioritize HR and Sales while investigating retention practices in R&D and Management.

6. Establish ongoing monitoring: Refresh the Power BI dashboard regularly and track attrition, satisfaction, promotion and workload indicators.

Management Takeaway

Employee attrition should be addressed through a targeted retention strategy rather than a single intervention. Priority areas are employee satisfaction, career progression, compensation and workload. Department-level action should consider both attrition rate and the number of employees leaving.

CRISP-DM at a Glance

CRISP-DM Phase	Project Output
Business Understanding	Define attrition and workforce-retention questions.
Data Understanding	Profile 14,999 employee records and workforce variables.
Data Preparation	Clean fields, standardize categories and create analytical measures/bands.
Modeling / Analysis	Build Power BI measures and comparative workforce visuals.
Evaluation	Interpret relationships between attrition and key workforce factors.
Deployment	Present executive findings and implement targeted retention actions.

Prepared as a Data Analytics / HR Employee Attrition Analytics Portfolio Project